DIDULANA LOKUGAMAGE

Computer Engineering Undergraduate at University of Peradeniya
Phone: +94778403272 | Email: didulanalokugamage@gmail.com | LinkedIn | GitHub | didulana.github.io

EDUCATION

2024 – Present **B.Sc. in Engineering (UG), Computer Engineering** — University of Peradeniya

2015 – 2022 **Physical Science (G.C.E. A/L)** — Rajapaksa Central College, Weeraketiya

- 9 A's — G.C.E. Ordinary Level Examination
- 3 A's — G.C.E. Advanced Level Examination

TECHNICAL PROJECTS

LexMed — Forensic Medicine Department Database System

Secure, role-based full-stack web application | MySQL, Node.js, Express.js, JavaScript, Bootstrap 5

- Designed a normalized MySQL database with role-based access control (JMO, department clerk, auditor roles) and append-only audit-log triggers to protect medico-legal records.
- Built a Node.js/Express REST API secured with JWT authentication, fully parameterized queries/stored procedures, and AES-256 encryption for sensitive identifiers such as patient NIC.
- Developed a role-aware frontend with a unified case timeline (MLEFs, PMRs, evidence) and Chart.js analytics dashboards for case reporting.

GitHub: LexMed-Forensic-Medicine-Department-Database-System

KidsRoute — School Transport Vehicle Management System

Real-time tracking and journey management platform | PostgreSQL, Express.js, React, Node.js, TypeScript, Socket.io, Leaflet

- A real-time school transport tracking and journey management platform that improves safety, transparency, and efficiency by connecting parents, drivers, and schools through live GPS tracking, automated attendance logging, and instant notifications.
- Designed and developed the Parent Dashboard (React, TypeScript, Tailwind CSS) with a live Leaflet map view and multi-child monitoring, and implemented the real-time GPS tracking logic using Socket.io for bidirectional WebSocket communication, powering live location updates and instant SOS broadcast alerts.
- Led the UI/UX design across the Parent, Driver, and Admin interfaces, managed the team's GitHub repository (branch strategy, pull request reviews, merge conflict resolution), and handled testing and deployment to ensure stable releases.

GitHub: e22-co2060-School-Van-Management-System

Fabric Defect Detection & Localisation Using Deep Neural Networks *(5th Place, Kaggle Competition)*

Computer vision & machine learning | Python, LightGBM, ResNet-18

- Developed an automated inspection system on the industrial AITEX fabric dataset (4096×256 px), placing 5th on the Kaggle leaderboard.
- Engineered domain-tailored texture features (Gabor wavelets, LBP, GLCM) and trained a LightGBM classifier achieving 90.17% accuracy (0.942 AUC).
- Benchmarked results against a fine-tuned ResNet-18 CNN (80.92% accuracy), showing that classical feature engineering can outperform deep transfer learning on limited industrial datasets.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, SQL, C / C++
Frontend Development: React, HTML5, CSS3, Tailwind CSS, Bootstrap 5
Backend & Real-Time: Node.js, Express.js, RESTful APIs, WebSockets (Socket.io)
Databases & Cloud/BaaS: PostgreSQL, MySQL, Supabase
Machine Learning & Data: PyTorch / OpenCV (or scikit-learn), LightGBM, Pandas, NumPy
Developer Tools & Workflow: Git, GitHub, Postman, Vite, Linux / Bash
UI/UX & Design Tools: Figma, Adobe Creative Suite

CERTIFICATIONS & COURSES

- Google AI Essentials — Google
- Web Design for Beginners — University of Moratuwa
- Python for Beginners — University of Moratuwa
- Leadership & People Management — Semrush Academy

PROFESSIONAL EXPERIENCE

2019 – Present **Founder & Creative Director** — Nix Studios
2023 – Present **Co-Founder & Photographer** — Evenraw

LEADERSHIP & ACTIVITIES

2025 – 2026 **Committee Member** — Association of Computer Engineering Students, UoP
2025 – Present **Co-Editor** — Engineering Students' Publication Society (GAUGE)
2024 – Present **Photographer** — Perabeats, University of Peradeniya
2024 – 2025 **Team Member – IGT / People Management** — AIESEC in Sri Lanka
2022 **Deputy Head Prefect** — Rajapaksa Central College, Weeraketiya
2022 **Head of the Technical Team** — Radapasa Media Unit
2016 – 2022 **Senior Member** — Radapasa Astronomical Society

AWARDS

- Silver Medal & Merit Award — Sri Lanka Junior Astronomy Olympiad
- Merit Award — Sri Lanka Physics Olympiad
- Finalist — PAD Awards by Pheonix Industries, ‘Brands for Life’ Category

REFERENCE

Nimesh Rubasinghe
Director, Neuroon Informatics
Phone: +94712515020